MODULE 05 / LESSON 01

Mathematics and Statistics for Machine Learning

Build the numerical language used to represent data, measure error, and improve a model.

DSACADEMY

WHY THIS MATTERS

Start from what you already know

Basic arithmetic, Python expressions, arrays, and averages

01 Explain vectors and matrices in plain language.

02 Use probability in a small worked example.

03 Complete the guided lab and verify gradients.

CORE CONCEPT

The vocabulary of this topic

Build the numerical language used to represent data, measure error, and improve a model.

vectors and matrices: A vector represents one ordered feature set; a matrix stacks vectors so one operation can transform many examples.

probability: Probability represents calibrated uncertainty about an outcome on a scale from zero to one.

loss functions: A loss function converts prediction error into a scalar objective whose definition determines what training rewards.

gradients: A gradient gives the local rate and direction of loss change for every trainable parameter.

FOUR CONNECTED IDEAS

How the concept works

01

vectors and matrices →

02

probability →

03

loss functions →

04

gradients

Matrix multiplication produces predictions; loss measures error; a gradient gives the direction of fastest local loss increase.

CORE CONCEPT

Read every part of the mechanism

Matrix multiplication produces predictions; loss measures error; a gradient gives the direction of fastest local loss increase.

vector: An ordered list of numbers.

matrix: A rectangular grid of numbers.

gradient: One derivative for each parameter.

WORKED EXAMPLE

Worked example by hand

For $x=[2,3]$, $w=[0.4,-0.1]$, and $b=0.2$, calculate $x\cdot w+b=0.7$, then compare with target 1.

01

Known values

02

Apply the mechanism

03

Interpret the result

RUN IT AND INSPECT THE RESULT

Map the concept to Python

```
import numpy as np
x = np.array([2.0, 3.0])
w = np.array([0.4, -0.1])
prediction = x @ w + 0.2
print(prediction)
```

OUTPUT

```
0.7
```

Each line corresponds to a concept already explained.

QUALITY GATE

Build the complete mental model

- [x] `(n,k) @ (k,m)` is valid because the shared dimension represents the same features.
- [x] Variance measures average squared distance from the mean; standard deviation restores original units.
- [x] A derivative is local slope; a gradient collects one local slope per parameter.
- [x] Gradient descent moves opposite the gradient by a distance controlled by the learning rate.

FOUR CONNECTED IDEAS

Notebook readiness map

01

vectors and matrices →

02

probability →

03

loss functions →

04

gradients

Use the concepts from this lesson to read and complete https://github.com/curiousily/AI-Bootcamp/blob/master/README.md.

RISK REVIEW

Common beginner mistakes

01 **Multiplying arrays without checking compatible dimensions.**

02 **Changing a parameter without connecting the update to loss.**

GUIDED LAB

Calculate vector shapes, a dot product, squared error, and one parameter update with visible intermediate values.

01

Predict

02

Run

03

Inspect

04

Explain

SESSION SYNTHESIS

Keep the system, not isolated definitions.

01 **Explain vectors and matrices in plain language.**

02 **Use probability in a small worked example.**

03 **Complete the guided lab and verify gradients.**

NEXT EVIDENCE

A mathematical notebook with hand calculations beside matching NumPy results.